

A REPORT

ON

PISTON FACTORY SIMULATION MODEL

BY

Names of the Students	ID. Nos
1. Aditya Agarwal	2022A4PS1207G
2. Sakshi Bharadwaj	2022A4PS1492H
3. Shreyansh Kedawat	2022A4PS0897P
4. Srujaan Sushant Parhate	2022A4PS0876H

AT

Commercial Consultancy Services (Flexsim) - Bangalore

A Practice School-I Station of

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

(July, 2024)

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1. Aditya Agarwal	2022A4PS1207G	B.E. Mechanical
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First and foremost, we would like to express our heartfelt thanks to Prof. Sujith R. in Charge, for his unwavering support and mentorship during this course of a month. He constantly guided us and acted as a bridge to resolve issues faced by us students. Prof. Sujith's constant encouragement, insightful feedback, and willingness to address our queries have been crucial in enhancing our skills and knowledge in the field. We are truly grateful for his invaluable guidance and mentorship.

We would also like to extend our deep gratitude to Mr. Arun KR and Mr. Arulraj A, our mentors, for their exceptional guidance and support throughout the internship. Their expertise in the field, along with their patience and willingness to share knowledge, has been truly inspiring. Their valuable insights, constructive feedback, and suggestions have been instrumental in refining our understanding of the project and ensuring its successful execution. We immensely thank Mr. Arun KR and Mr. Arulraj A for their constant motivation and belief in our abilities.

We would also like to extend our heartfelt gratitude to the PS division at BITS Pilani for this opportunity to get industrial exposure. In addition, we express our token of gratitude to the entire **Commercial Consultancy Services(Flexsim)** team for having us onboard and guiding us. At last, we would like to thank our fellow students at the station for the constructive discussions and support provided during the course of one month.

**BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE PILANI
(RAJASTHAN) Practice School Division**

Station: Commercial Consultancy Services (Flexsim) **Centre:** Bengaluru

Duration: 2 months **Date of Start:** 28/05/24

Date of Submission: 22/07/24

Title of the Project: Piston Factory Simulation Model

ID No.(s)	Name(s)	Discipline(s)
2022A4PS1207G	Aditya Agarwal	Mechanical
2022A4PS1492H	Sakshi Bharadwaj	Mechanical
2022A4PS0897P	Shreyansh Kedawat	Mechanical
2022A4PS0876H	Srujaan Sushant Parhate	Mechanical

**Name(s) and designation(s)
of the expert(s):** Mr. Arun KR and Mr. Arulraj A

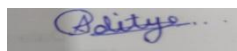
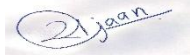
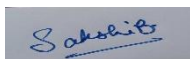
Name(s) of the PS Faculty: Mr. Sujith R.

Key Words: bottleneck, entities , operator . optimization , queue , simulation

Project Areas: finding bottle neck and optimization of the simulation model

Abstract:

This project focuses on the analysis of piston simulation model to figure out the machine and output utilization , bottleneck in the assembly and all the possible ways to increase its efficiency.



Signature(s) of Student(s)

Signature of PS Faculty

Date : 22/07/24

Date

TABLE OF CONTENTS

SL No.	NAME	PAGE No.
1	Introduction	1
2	Main Text	3
2.1	Progress Meet 1	4
2.2	Progress Meet 2	10
2.3	Final Progress Meet	14
3	Suggestions	18
4	Summary	19
5	Conclusion	20
6	Reference	21
7	Glossary	22

INTRODUCTION

FlexSim is simulation software used **to model operations such as** manufacturing, **transportation** and healthcare. It **provides** 3D visualization, modeling tools for creating detailed models, simulation capabilities and **optional APIs for performance analysis**. It is widely used to **improve** processes and make informed decisions by simulating **real-life situations**.

Flexsim is basically a **Discrete Event Simulation Software** which simulates systems which change in **State** at discrete points in time as a result of specific **Events**.

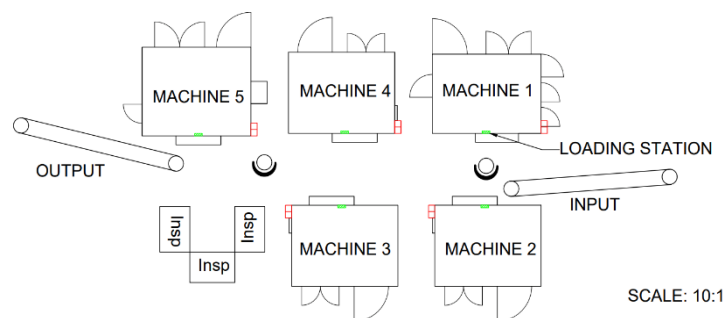
Some common uses of Simulation are to increase efficiency, reducing waiting time and queue sizes etc. Simulation helps planners to capture things which excel sheets don't.

According to what we have learned till now, the 4 major steps of making a 3D simulation are,

1. Build a 3D model
2. Get Data from a 3D model
3. Build a Process Flow Model
4. Link the Models

After completing all the steps we can compare several models to remove the bottlenecks and increase the efficiency to find the best solution.

In our case we were given a **Piston Factory Simulation** where our goal was to find out the bottleneck and tell which process flow will yield the best output.



Using the given AutoCAD layout of the factory we simulate the 3D model of the Piston Factory and after inputting the given input data we get the statistics of the model in the dashboard which can be used to make further advancements.

Overall, this report **highlights** the **importance** of **changing apprenticeships**. **Simulating** a piston manufacturing **plant using FlexSim software provides** insights, recommendations, and a deeper understanding of **business processes**. These findings contribute to **business** process optimization and **demonstrate** the effectiveness of simulation technology in **improving** operational **efficiency** and **promoting innovation**.

MAINT TEXT

Our aim is to analyze the assembly cell of the Piston factory to figure out the throughput, machines utilization and operators' utilization. To find out the bottleneck and to increase the line output by possible ways, the input data being:

Input Data:

Number of machines: 5

Number of man: 1

Loading time: 6 sec

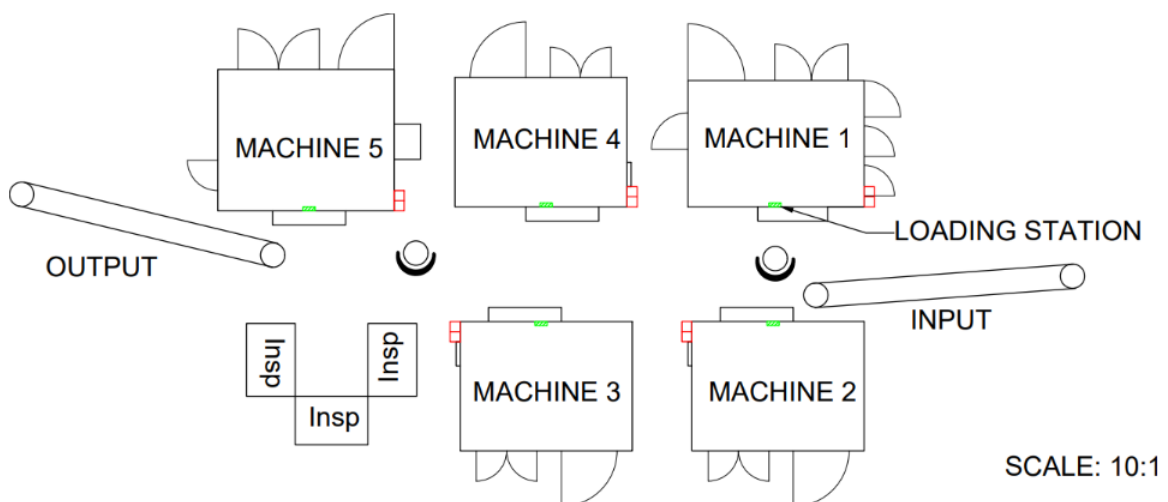
Unloading time: 4 sec

Machining Time (Processing Time): 68 sec

Max carrying capacity of man: 2

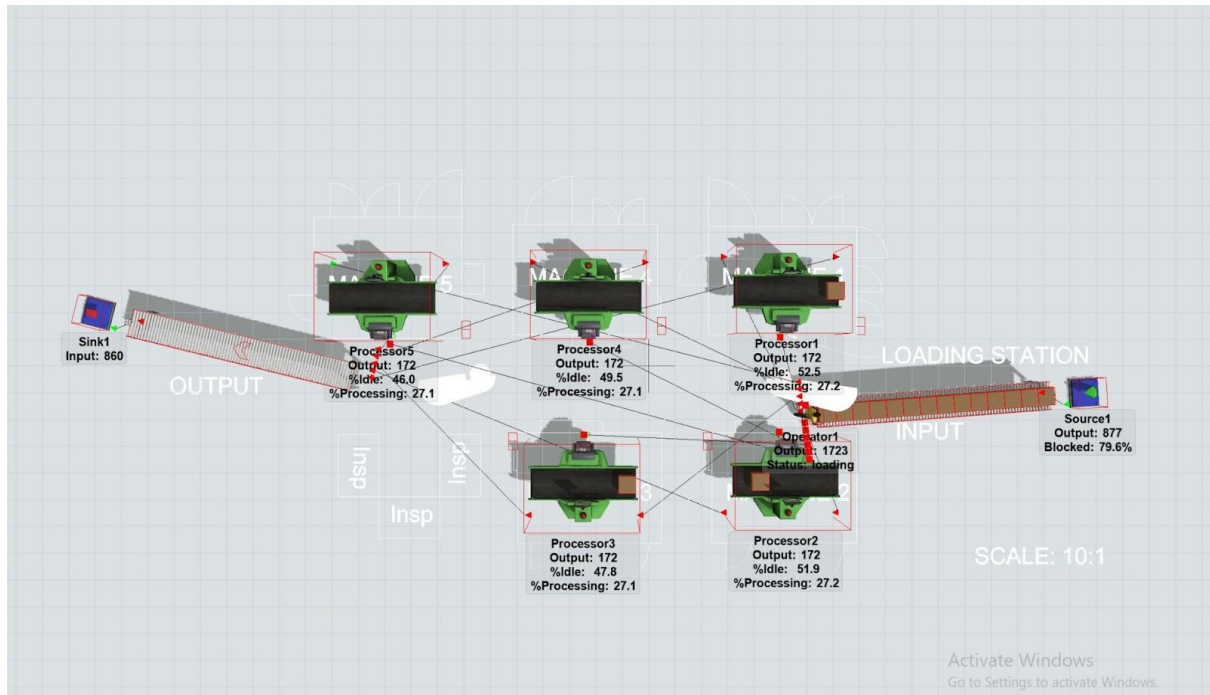
Max walking speed: 1.33 m/s

AutoCAD Layout:



Progress meet - 1

1st Case with one operator:



Analysis

Input=877

Output=860

PROCESSOR	OUTPUT	%IDLE
1	172	52.5
2	172	51.9
3	172	47.8
4	172	49.5
5	172	46

Total output of processors = $172+172+172+172+172$
= 860

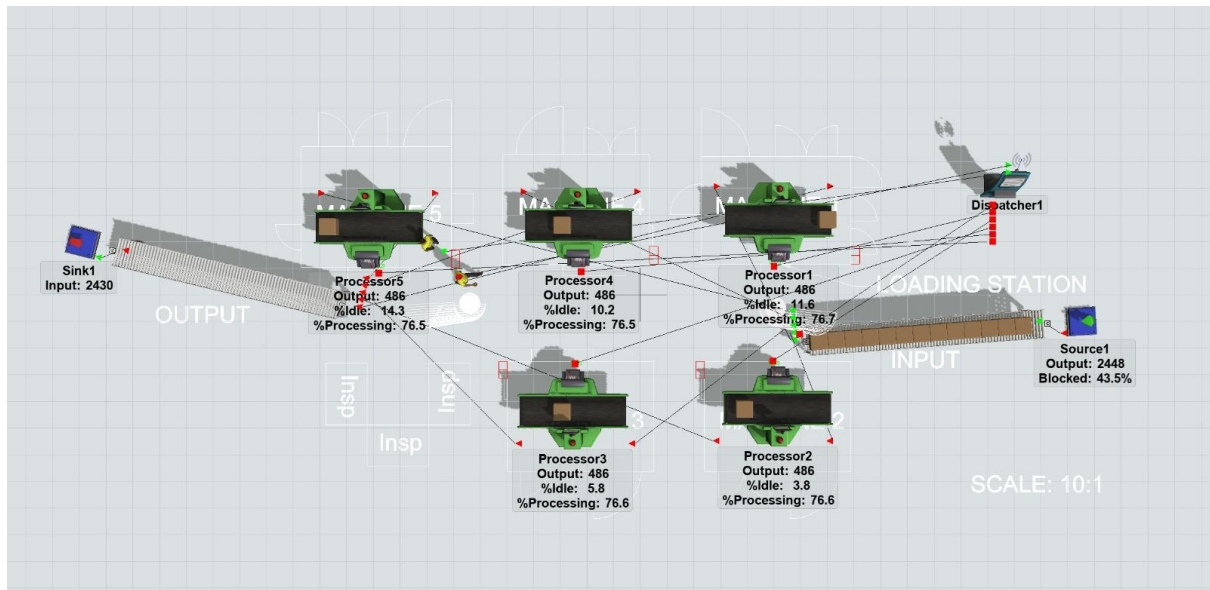
Difference = $877-860 = 17$

Efficiency=98.06%

Average percentage idle= 49.54%

There was a **discrepancy of 17 flow items**, and the processors were approximately **idle for 50%** of the time. Therefore, we proceeded with another case of 2 operators.

2nd Case with Two Operators-



Analysis

Input = 2448

Output = 2430

PROCESSOR	OUTPUT	%IDLE
1	486	11.6
2	486	3.8
3	486	5.8
4	486	10.2
5	486	14,3

Total output of processors = $486+486+486+486+486$
 $=2430$

Difference = $2448-2430 = 18$

Efficiency = 99.26%

Average percentage idle = 9.14%

There was a **discrepancy of 18 flow** items and also the processors were approximately **idle for 9%** of the time.

Comparison between 1st and 2nd case:

Sl. No.	Parameter	With 1 operator	With 2 operator	Percentage difference
1.	Input	877	2448	179.13%
2.	Output	860	2430	182.55%
3.	%idle	49.54	9,14	81.55% (decrease)

The results were better in the second case.

But there were still some issues with the model that were decreasing efficiency, such as:

Despite having the capacity for 2 boxes, the operator sometimes carries only 1 box.

The operator carries 1 output from the processor and 1 flow item from the queue to the final conveyor belt rather than putting unprocessed flow item into the processor and then taking the processed flow item

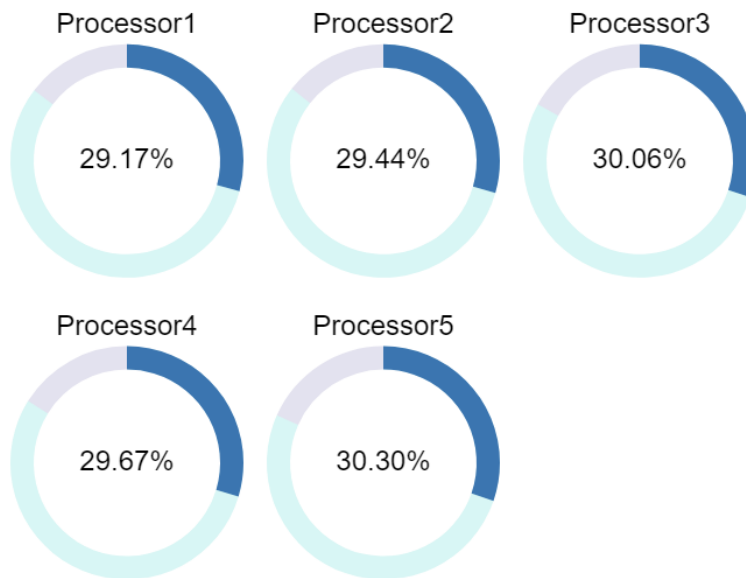
When the operator takes two flow items from the queue, he first unloads one of them into an empty processor. With one flow item left, he moves to another processor that has finished its task. He then loads the processed flow item and unloads it onto the output conveyor. After that, he returns to the empty processor to unload the remaining unprocessed item and then goes back to the queue.

The primary feedback we received on our project was to utilize a global list, implement a process flow, and make use of a dashboard to address the issues we are encountering.

Based on the feedback we received, we have incorporated a dashboard into our simulation. Furthermore, we have added a queue in our layout.

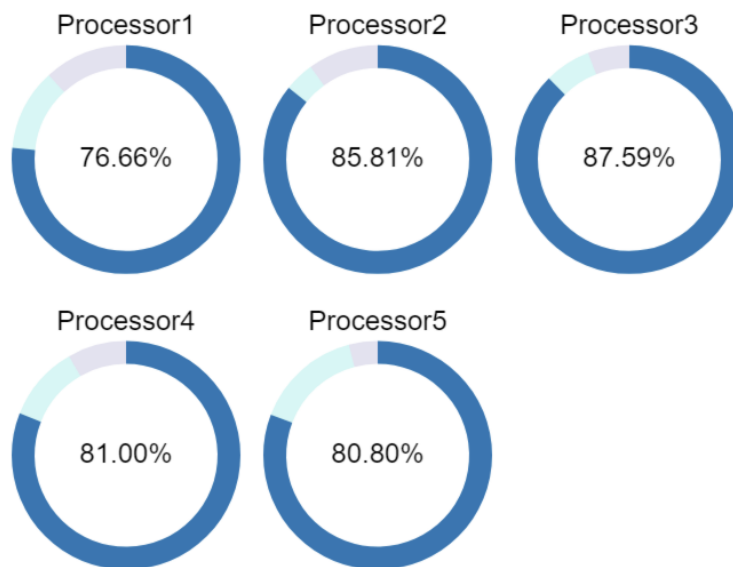
Processor States with 1 Operator

■ Processing ■ Idle ■ Waiting for transporter



Processor States with 2 Operators

■ Processing ■ Idle ■ Waiting for transporter

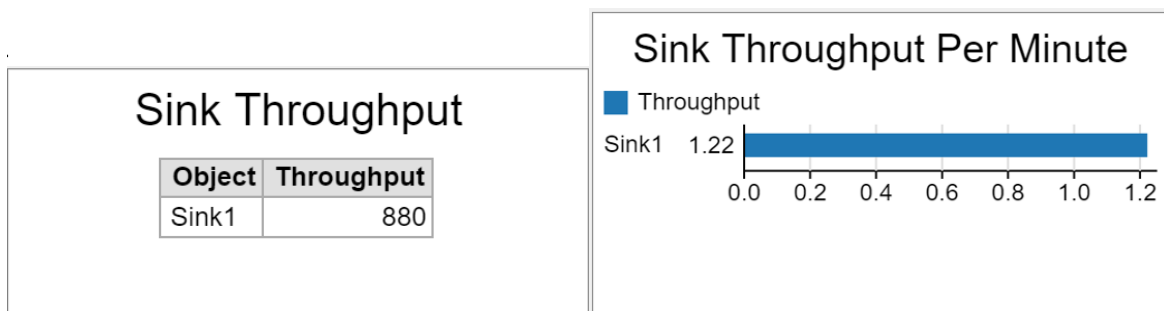


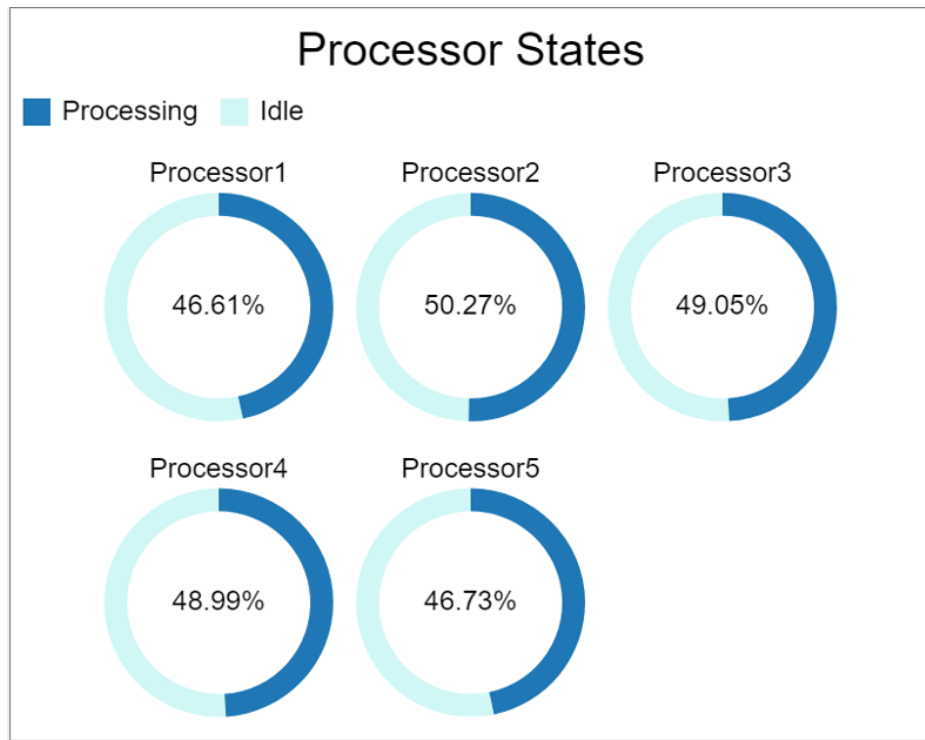
Progress meet - 2

After thoroughly reviewing the instructor's suggestions, we implemented several changes to the model. Specifically, we utilized process flow, lists, and queues. Based on these modifications, we developed two new cases.

Case1:

The operator goes to the final queue after the flow items are processed and then return to the initial queue.





Although we achieved an improved result, we did not reach the optimal solution, as there were instances when the processors were idle and the operator was not engaged in any tasks. To address this issue, we devised a second approach.

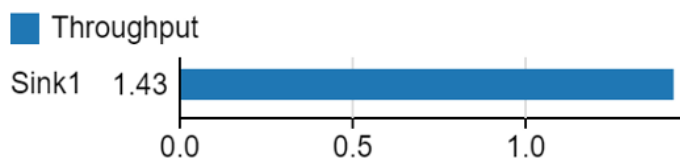
Case 2:

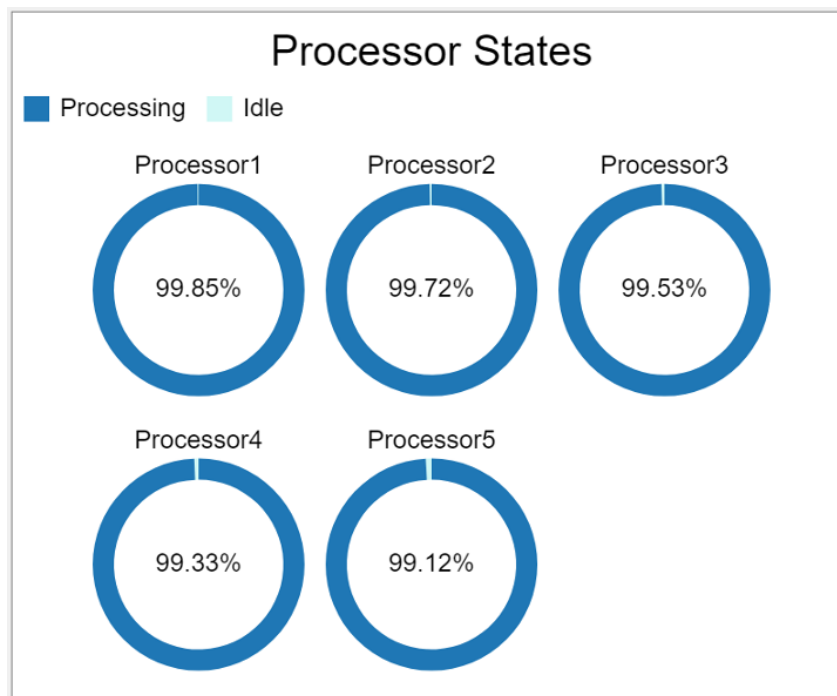
The operator proceeds to the Initial Queue after unloading flow items to the processors and loading unfinished flow items. The operator then moves to the processors, where the flow items are fully processed, and exchanges them with the unprocessed ones. This process continues until all unprocessed flow items are loaded into the processors. Finally, the operator moves to the Final Queue to retrieve all the processed items.

Sink Throughput

Object	Throughput
Sink1	1028

Sink Throughput Per Minute

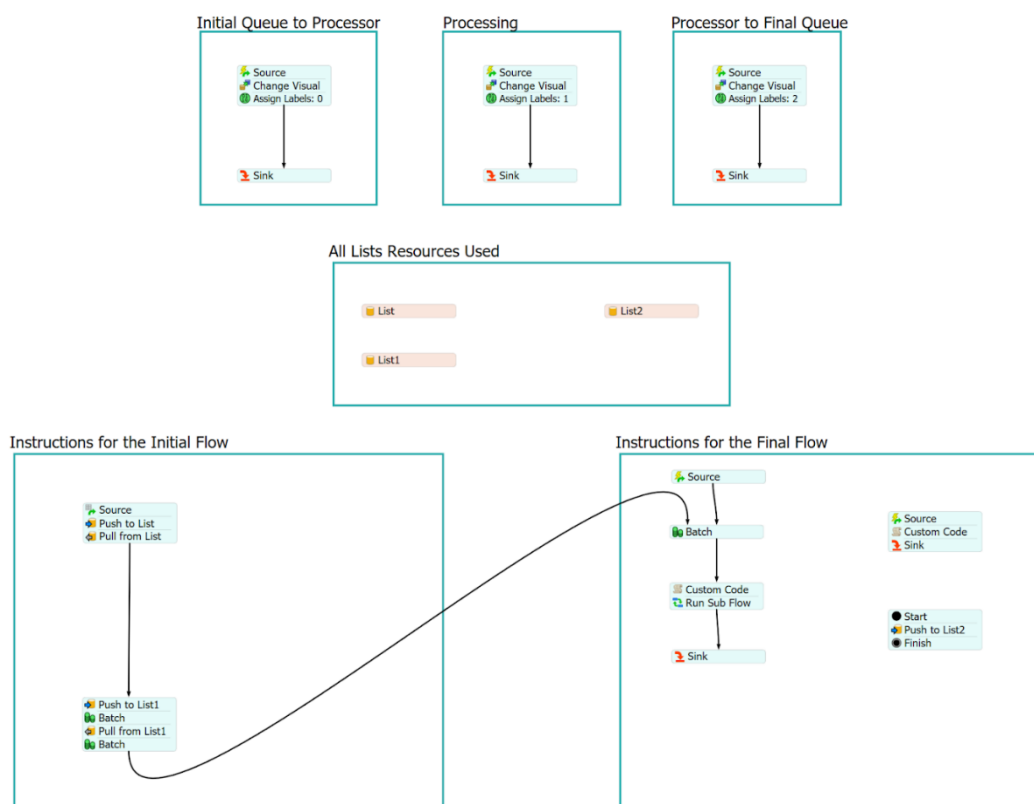


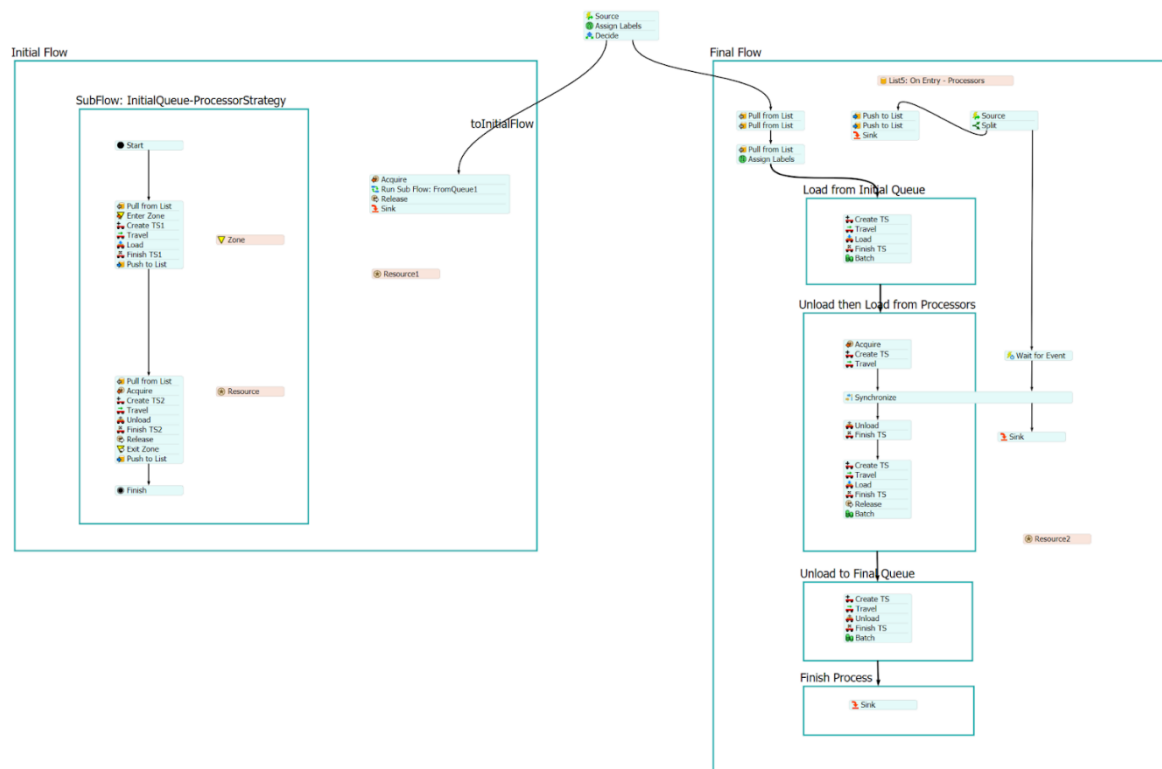


Based on our model, several suggestions from industry professional and mentor were provided. They recommended adding a parameter list to enable the modification of input data and the operator's properties, such as carrying capacity and speed. Additionally, they advised incorporating a dashboard to display sink throughput per minute. It was also noted that the operator was not following the shortest path, resulting in inefficiencies due to excessive travel time between processors.

Final Presentation:

Based on the suggestions provided, we optimized our model by incorporating custom FlexScript code to implement various functions. These functions include calculating the shortest distance and using permutations and combinations to determine the optimal sequence for the operator to retrieve flow items from the processors.





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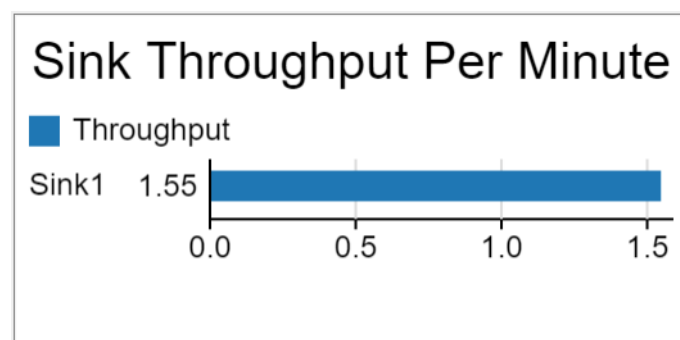
FlowItem Bin | Custom Code - Custom Code
1 /**Custom Code*/
2
3 Object current = param(1);
4 treenode activity = param(2);
5 Token token = param(3);
6 treenode processFlow = ownerobject(activity);
7
8 //Operator Parameters-->
9 int k = Model.parameters.k; //2
10 int oprLoadTime = Model.parameters.LoadTime;
11 Object op = Model.find("Operator1");
12 Object nn = Model.find("NN1");
13 double oprSpeed = Model.parameters.MaxSpeed;
14 double Timequeue1 = ((oprSpeed/2) + (distancetottravel(op,nn)/oprSpeed)) + Model.time + (oprLoadTime)*k;
15
16 // Function to generate and print all permutations of all combinations-->
17 Array arr = [1,2,3,4,5];
18 Array allPermutations;
19 generateAllPermutationsOfCombinations(arr, k, allPermutations);
20
21
22 //Time Parameters -->
23 double minTime = 43200; //in seconds ( 12 hours ) = model run time
24 double Time = 0;
25 int minidx = 1;
26
27 //For loop to get minimum Time -->
28 for(int i = 1; i<=allPermutations.length; i++){
29     Time = boxTime(allPermutations[i], Timequeue1);
30     // print(Time+ Tql, allPermutations[i]);
31     if(Time < minTime){
32         minTime = Time;
33         minidx = i;
34     }
35 }
36
37 Group group = Group("processorGroup");
38 token.labels.assert("toProcessorArr").value = allPermutations[minidx];
39
40 //Print the minimum selected path-->
41 print(minidx);
42 print(allPermutations[minidx]);

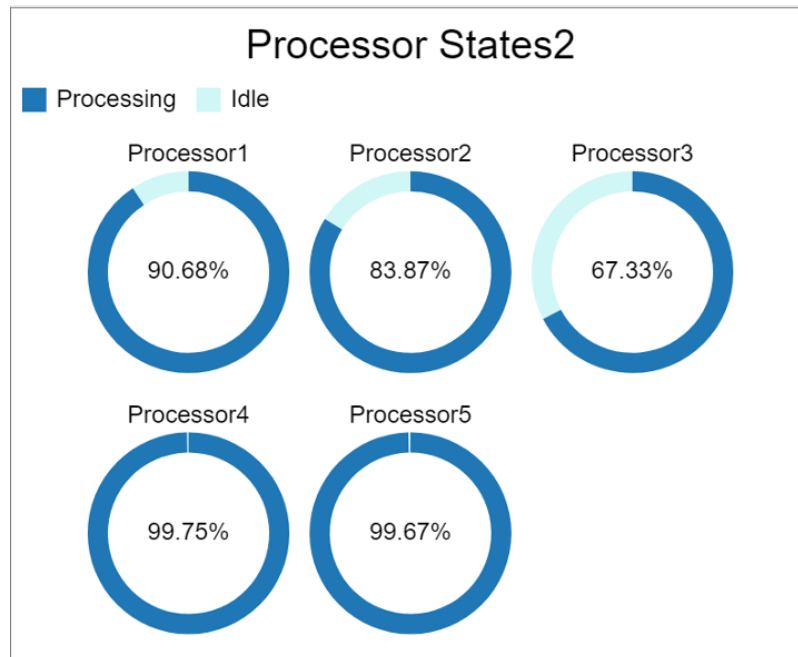
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Apply All Apply OK Cancel

Sink Throughput

Object	Throughput
Sink1	1114





Based on our final model, we determined that the output was higher than in any previous case. However, we identified a bottleneck: only two processors were in continuous use, leaving the remaining three processors idle for a significant portion of time.

Suggestions we got from Arul sir

1. To address the bottleneck, we could increase the number of operators to ensure all processors are utilized. Additionally, we could assign specific tasks to each operator. For instance, one operator could be responsible for loading the processors, while another could handle unloading, thereby reducing travel time significantly.

2. In practical applications, it is crucial to make dashboards more effective and user-friendly. When collaborating with industry professionals who may lack technical knowledge of the model's intricacies, it is essential to present the data in a clear and comprehensible manner. This approach helps to demonstrate convincingly that the model predicts maximum possible efficiency under the given conditions, thereby facilitating better understanding and acceptance of the model's outcomes.

SUMMARY

The Piston Simulation Model project in FlexSim involves creating a virtual representation of a piston mechanism to analyze its performance, identify potential improvements, and optimize its design and operation. FlexSim, a 3D simulation software, is used to build and test this model, allowing for detailed analysis and visualization of the piston system in a virtual environment.

Our aim is to analyze the assembly cell of the Piston factory to figure out the throughput, machines utilization and operators' utilization. To find out the bottleneck and increase the line output by different possible ways.

CONCLUSION

Our aim in the piston factory simulation model project using Flexsim software was to improve operations and identify bottlenecks. We have achieved important things throughout the project and we were constantly working to improve our standards. After initially running simulations using one and two workers, we concluded that the model using two workers was better.

However, there were still many problems in the system that need to be solved. The global list helped us track various points in the simulation, while the process flow allows us to see and optimize the order of operations.

Digging deeper into the capabilities of Flexsim, we explore other optimization methods to identify and solve physical problems. Continuous learning and application of these techniques is crucial to improving our standards. This addition greatly optimizes the overall efficiency of the system.

The process flow allows us to manage workflow more effectively and make work more efficient. To further improve the process, we implemented custom code into the process that guides the operator to pick up the nearest box from the sink. This change reduces travel time and increases the overall efficiency of the system.

After implementing everything we learnt during this internship, we observed that only 2 processors were being used to near full efficiency when only one operator was employed. In our case, this was the bottleneck.

We can say that the simulation, when run to its utmost efficiency with the given input data, only utilizes 2 processors at their full capacity. And to solve this bottleneck we can employ more operators. In this case using more operators with the given process will yield maximum efficiency. Or we can add one more operator and divide the tasks of picking up the box from the input sink and putting the box in the output sink. This will ultimately solve the bottleneck problem of this simulation.

In conclusion, we are very grateful for the opportunity to learn and use Flexsim software during this project. These experiences have improved our understanding of simulation modeling and optimization, providing us with valuable skills and knowledge that will be useful in our future work. We appreciate the guidance and support we have received throughout this process and look forward to using this information to improve the efficiency and effectiveness of our business.

REFERENCES

Flexsim website: <https://www.flexsim.com/>

Flexsim tutorials:

<https://docs.flexsim.com/en/24.1/Tutorials/FlexSimBasics/BasicsOverview/BasicsOverview.html>

https://youtu.be/YJgJzyDHZ8o?si=1oGSBENMhuU_ihEp

GLOSSARY

FlexSim: A powerful simulation software used for modelling, analysing, and optimizing systems in various industries.

Simulation Model: A digital representation of a real-world system created within FlexSim to study its behaviour and performance.

Bottleneck: A stage in the process that limits the overall system capacity due to its lower throughput compared to other stages.

Cycle Time: The total time taken for an entity to move through the entire process from start to finish.

Idle Time: The period when a resource is not in use, often due to waiting for other processes to complete.

Entity: An item or unit (e.g., product, task) that moves through the system in the simulation model.

Discrete Event Simulation: A type of simulation where the operation of a system is represented as a chronological sequence of events.

Performance Metrics: Quantitative measures used to assess the performance of the system, such as average queue length, wait times, and resource utilization.

Optimization: The process of making changes to the system or model to improve its performance, efficiency, and effectiveness.

Process Simulation: Using a simulation model to replicate and study the behaviour of a real-world process.

Data Collection: Gathering information on system performance, resource usage, and other relevant metrics during the simulation.

Output Analysis: Examining the results generated by the simulation to draw conclusions and make informed decisions.

Queue: A line or waiting area where entities are held until resources become available to process them.

Discrete Event Modeling: Creating a simulation model that focuses on events occurring at specific points in time to drive the system's dynamics.